

Real-Time Inflation Expectations: Households vs. Generative AI*

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Abstract

We compare real-time inflation expectations between households and generative AI using identical survey settings. We find that generative AI exhibits systematically downward-biased inflation expectations with significantly lower variation than households. These results suggest important differences in how households and AI process inflation information.

JEL Classification: C90; D84; E27; E31; E58; E71

Keywords: Generative AI; Inflation expectations; Information provision;
Large language models; Survey experiment

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Appendix A. Questionnaire

Q.1. In your opinion, do you think prices have gone up or down compared with a year ago? (By “prices” we mean the overall level of prices of the goods and services you purchase, including consumption tax.)

- (1) Have gone up significantly
- (2) Have gone up slightly
- (3) Have been unchanged
- (4) Have gone down slightly
- (5) Have gone down significantly

Q.2. In your opinion, how much do you think prices have changed year-on-year compared with a year ago?

- | | |
|---------------------------|--------------------|
| (1) Around +12% or higher | (+11.5% or higher) |
| (2) Around +11% | (+10.5% to +11.4%) |
| (3) Around +10% | (+9.5% to +10.4%) |
| (4) Around +9% | (+8.5% to +9.4%) |
| (5) Around +8% | (+7.5% to +8.4%) |
| (6) Around +7% | (+6.5% to +7.4%) |
| (7) Around +6% | (+5.5% to +6.4%) |
| (8) Around +5% | (+4.5% to +5.4%) |
| (9) Around +4% | (+3.5% to +4.4%) |
| (10) Around +3% | (+2.5% to +3.4%) |
| (11) Around +2% | (+1.5% to +2.4%) |
| (12) Around +1% | (+0.5% to +1.4%) |
| (13) Around +0% | (−0.5% to +0.4%) |
| (14) Around −1% | (−1.5% to −0.6%) |
| (15) Around −2% or lower | (−1.6% or lower) |

Q.3. What do you think the consumption tax rate on food will be from April next year?

- (1) 0%
- (2) 1%
- (3) 2%
- (4) 3%
- (5) 4%
- (6) 5%

- (7) 6%
- (8) 7%
- (9) 8%
- (10) 9%
- (11) 10%

Q.4. Over the next year, do you think prices will go up or down?

- (1) Will go up significantly
- (2) Will go up slightly
- (3) Will be unchanged
- (4) Will go down slightly
- (5) Will go down significantly

Q.5. Over the next year, how likely do you think that changes (1)–(7) below will occur in the rate of price increase? Please enter a number for each. The numbers should total 100.

In this question, you will think about the percent chance of something happening in the future. 0 means there is absolutely no chance, and 100 means that it is absolutely certain. For example:

2–5%: “almost no chance”

around 18%: “not much chance”

47–52%: “a pretty even chance”

around 83%: “a very good chance”

95–98%: “almost certain”

- | | |
|-------------------------------------|---------|
| (1) increase by 12% or more | _____ % |
| (2) increase by 8% to less than 12% | _____ % |
| (3) increase by 4% to less than 8% | _____ % |
| (4) increase by 2% to less than 4% | _____ % |
| (5) increase by 0% to less than 2% | _____ % |
| (6) decrease by 0% to less than 2% | _____ % |
| (7) decrease by 2% or more | _____ % |

Q.6. Please read the information below.

Group B (placebo): “According to estimates by the Ministry of Internal Affairs and Communications, Japan’s population as of May this year had decreased by 0.4% compared with the previous year.”

Group C (treatment 1): “The Bank of Japan expects prices, excluding fresh food, to increase by 2.8% compared with the previous year in the current fiscal year.”

Group D (treatment 2): “The Bank of Japan has set its price stability target at a year-on-year rate of increase of 2%.”

Q.7. Group B: According to estimates by the Ministry of Internal Affairs and Communications, how much had Japan’s population as of May this year changed compared with the previous year?

- (1) decrease of 1.4%
- (2) decrease of 0.4%
- (3) increase of 0.4%
- (4) increase of 1.4%

Group C: By how much did the Bank of Japan expect prices excluding fresh food to rise this fiscal year compared with the previous year?

- (1) increase by 1.8%
- (2) increase by 2.8%
- (3) increase by 3.8%
- (4) increase by 4.8%

Group D: At what year-on-year rate of increase has the Bank of Japan set its price stability target?

- (1) 0%
- (2) 1%
- (3) 2%
- (4) 3%

Q.8. I would like to ask again about the price outlook. Over the next year, how much do you think prices will change year-on-year?

(1) Around +12% or higher	(+11.5% or higher)
(2) Around +11%	(+10.5% to +11.4%)
(3) Around +10%	(+9.5% to +10.4%)
(4) Around +9%	(+8.5% to +9.4%)
(5) Around +8%	(+7.5% to +8.4%)
(6) Around +7%	(+6.5% to +7.4%)
(7) Around +6%	(+5.5% to +6.4%)
(8) Around +5%	(+4.5% to +5.4%)
(9) Around +4%	(+3.5% to +4.4%)
(10) Around +3%	(+2.5% to +3.4%)
(11) Around +2%	(+1.5% to +2.4%)
(12) Around +1%	(+0.5% to +1.4%)
(13) Around +0%	(−0.5% to +0.4%)
(14) Around −1%	(−1.5% to −0.6%)
(15) Around −2% or lower	(−1.6% or lower)

Q.9. Please indicate your gender.

- (1) Male
- (2) Female
- (3) Prefer not to answer

Q.10. Please enter your age.

_____ years old

Q.11. Please indicate where you live. [Drop-down menu; the placeholder reads “Please select.”]

- (1) Hokkaido
- (2) Aomori
- ~
- (47) Okinawa
- (99) Others

Q.12. Are you married?

- (1) Not married (never married, divorced, or widowed)
- (2) Married

Q.13. Please indicate your occupation.

- (1) Company employee / officer
- (2) Self-employed
- (3) Professionals (doctors, lawyers, hairdressers, designers, etc.)
- (4) Civil servant
- (5) Student
- (6) Housewife / househusband
- (7) Part-time workers and freelancers
- (8) Unemployed / retired
- (9) Others

Q.14. Please indicate your educational background.

- (1) Primary and secondary school graduate
- (2) High school graduate
- (3) Technical college graduate
- (4) Vocational school graduate
- (5) Junior college graduate
- (6) University graduate
- (7) Graduate-school graduate
- (8) Studying at or enrolled in school

Q.15. Please indicate your household's total annual take-home income (total income excluding taxes and social contributions from January to December of the previous year).

- (1) Less than 2 million yen
- (2) 2 million to less than 3 million yen
- (3) 3 million to less than 4 million yen
- (4) 4 million to less than 5.5 million yen
- (5) 5.5 million to less than 7.5 million yen
- (6) 7.5 million to less than 9.5 million yen
- (7) 9.5 million to less than 12 million yen
- (8) 12 million yen or more